

Macau story



Macau story: Simplified



Suppose

- We toss the coin 10 times and get 8 tails

Has the Casino rigged the coin?

Macau story



Suppose

- The probability that a coin returns tail is 0.5
- We toss the coin 10 times and get 8 tails

What is the probability of getting 8 tails in 10 tosses?

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What is the probability of getting 8 tails in 10 tosses?

- Answer:

$$P(T = 8) = \binom{N}{T} \cdot 0.5^T \cdot (1 - 0.5)^{N-T} = 45/1024$$

$$\approx 0.044 \quad (\text{around } 4.4\%)$$

More likely to be rigged?

Introducing competing hypotheses



H_{fair} : The coin is fair

H_{rigged} : The coin is rigged

Probability of observing the data (T tails in N tosses) under each hypothesis:

$$P(T|H_{\text{fair}}, N) = \binom{N}{T} \cdot (0.5)^T \cdot (1 - 0.5)^{N-T}$$

$$P(T|H_{\text{rigged}}, q, N) = \binom{N}{T} \cdot q^T \cdot (1 - q)^{N-T}$$

T is the number of tails; coin bias q is the tail probability under H_{rigged} .

Under which hypothesis is the data more likely?

- Uniform prior on q : $P(q|H_{\text{rigged}}) = 1$ on $[0, 1]$

Bayes factor



Bayes factor:

$$B_f^r = \frac{P(T|H_{\text{rigged}}, N)}{P(T|H_{\text{fair}}, N)}$$

Evidence under rigged hypothesis:

$$P(T|H_{\text{rigged}}, N) = \int_0^1 P(T|H_{\text{rigged}}, q, N) \cdot P(q|H_{\text{rigged}}) dq$$

Uniform prior on q : $P(q|H_{\text{rigged}}) = 1$ on $[0, 1]$

$$P(T|H_{\text{rigged}}, N) = \frac{1}{N+1}$$

$$B_f^r \approx 2$$

Is the coin rigged though?



The Bayes factor is two. What does this mean?

- The data is twice as likely under the rigged hypothesis than under the fair hypothesis.

Is this strong evidence for the rigged hypothesis?

- Not really.
- For example, repeating the experiment 10 times would likely yield at least one experiment with a Bayes factor of 2 or more, even if the coin is fair.

Repeating the experiment 7 times



You repeat the experiment 7 times, each experiment gives 8 tails in 10 tosses.
- Is the coin rigged?

Surely it's rigged now?



You have seven datasets $\vec{d}$

- Let us assume that the datasets are independent

$$P(\vec{d}|H_{\text{rigged}}, N) = \prod_{i=1}^7 P(d_i|H_{\text{rigged}}, N)$$

$$P(\vec{d}|H_{\text{fair}}, N) = \prod_{i=1}^7 P(d_i|H_{\text{fair}}, N)$$

$$B_f^r \approx (2.069 \dots)^7 \approx 162$$

- Is the coin rigged now?

Surely it's rigged now?

— Are Casino Games Rigged?

Posted on Wednesday, May 14, 2025 by Ajush | Questions & Answers

Are Casino Games Rigged or Fair?

The Truth Behind Casino Game Mechanics

Ever walked away from a slot machine thinking, "No way that wasn't fixed"? You're not alone. It's a question that lingers in the minds of countless gamblers: are casino games rigged? It's a fair concern, considering the money involved, the rapid pace of wins and losses, and the sheer mystique surrounding these glittering temples of chance. So let's pull back the velvet curtain and peek into the gears of the games.

At the heart of every legit casino game lies mathematical structure, algorithmic precision, and regulatory oversight. It's not a magic show — it's a numbers game. Each game is programmed to offer the house a mathematical advantage (more on that later), but that doesn't mean it's unfair. Think of it like playing basketball on a court where the rim is just slightly tilted — you can score, but it's not exactly even.

Why Most Casino Games Aren't Truly "Rigged"

Let's settle this upfront: reputable casinos don't need to rig games — the house edge does that work for them. If a casino has properly programmed games with a built-in advantage, they already win over time. Manipulating outcomes further would not only be redundant, but also wildly illegal in licensed jurisdictions.

Casinos are heavily monitored. Rigging games could mean losing their license, getting slapped with million-dollar fines, and going out of business. That's a big risk to take when your games are already statistically profitable.

Sure, you might hit a cold streak on the roulette wheel or feel like that blackjack dealer always pulls a 21 when you're sitting pretty at 20. But that's randomness at play, not rigging.

You repeat the experiment 7 times, each experiment gives 8 tails in 10 tosses.

- Is the coin rigged?
- A rigged coin is around 162 times more likely to generate the data than a fair coin.

It's still probably not rigged. Why?

- See the news story on the left

Posterior odds

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Bayes factor is not the end of the story.

- We also need to consider the prior odds:

$$O_f^r = \frac{P(H_{\text{rigged}} | \vec{d}, I)}{P(H_{\text{fair}} | \vec{d}, I)}$$

$$= \frac{P(\vec{d} | H_{\text{rigged}}, I)}{P(\vec{d} | H_{\text{fair}}, I)} \cdot \frac{P(H_{\text{rigged}} | I)}{P(H_{\text{fair}} | I)} = 162 \cdot \frac{P(H_{\text{rigged}} | I)}{P(H_{\text{fair}} | I)}$$

Only < 1 in 10,000 casinos are rigged

- Macau Casinos are heavily regulated.

- Prior odds of rigged vs fair: 1/10,000

$$O_f^r = 162 \cdot \frac{1}{10,000} = 0.0162$$

Conclusion: The coin is still probably not rigged.

Home casino



Home casino

- You hear from around 50% of your friends who plays at the casino that the casino cheats.

You get 8 tails in 10 tosses, 7 times in a row.

- Is the coin rigged now?

$$\begin{aligned} O_f^r &= \frac{P(H_{\text{rigged}} | \vec{d}, I)}{P(H_{\text{fair}} | \vec{d}, I)} \\ &= \frac{P(\vec{d} | H_{\text{rigged}}, I)}{P(\vec{d} | H_{\text{fair}}, I)} \cdot \frac{P(H_{\text{rigged}} | I)}{P(H_{\text{fair}} | I)} = 162 \cdot \frac{P(H_{\text{rigged}} | I)}{P(H_{\text{fair}} | I)} \end{aligned}$$

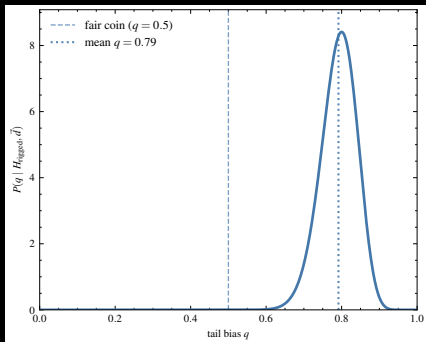
Prior odds of rigged vs fair: 1/1

$$O_f^r = 162 \cdot \frac{1}{1} = 162$$

Conclusion: The coin is probably rigged.

What can we tell about the coin?

What can we tell about the coin?



$$P(q|H_{\text{rigged}}, \vec{d}) = \frac{P(\vec{d}|H_{\text{rigged}}, q) \cdot P(q|H_{\text{rigged}})}{P(\vec{d}|H_{\text{rigged}})}$$

- Uniform prior: $P(q|H_{\text{rigged}}) = 1$ on $[0, 1]$

$$\begin{aligned} &= \frac{\prod_{i=1}^7 P(d_i|H_{\text{rigged}}, q) \cdot P(q|H_{\text{rigged}})}{P(\vec{d}|H_{\text{rigged}})} \\ &= \frac{\prod_{i=1}^7 \binom{N}{T} \cdot q^T \cdot (1-q)^{N-T} \cdot P(q|H_{\text{rigged}})}{P(\vec{d}|H_{\text{rigged}})} \end{aligned}$$

The coin is probably rigged and returns tails around 70-90 percent of the time.

Summary



Summary

- Bayes factor is the ratio of the likelihood of the data under two competing hypotheses.
- Bayes factor alone is not enough to determine which hypothesis is more likely. We also need to consider the prior odds.
- The posterior odds is the product of the Bayes factor and the prior odds.
- The coin in Macau is probably not rigged, but a coin in a home casino with similar data is probably rigged.
- We can also estimate the bias of the coin using the posterior distribution of q .

Open question: Detection of lensing

Suppose that a gravitational-wave signal is detected

- We analyse the signal under "microlensing" and "regular" hypotheses

Bayes factor is around 100

Is the signal microlensed?

- What are the factors that we need to consider to answer this question?